[reason: $\nabla(J_k - J_{Euc}) = \underbrace{(\nabla - \nabla_k)}_{\sim \nabla g_k} J_k$

$\Rightarrow \|\nabla(J_k - J_0)\|_{C^\alpha_{g_{Euc}}} \leq \Psi(k^{-1}) \|J_k\|_{C^\alpha_{g_{Euc}}} \leq \Psi(k^{-1})$]

<u>Claim</u> For $k \gg 1$ we have holo. charts $\{z^{(k)}_1, \dots, z^{(k)}_n\}$ on $B(0, R/2)$ w.r.t. J_k cx-structure, s.t.

$$\|z^{(k)} - z\|_{C^{2,\alpha}(B(0,R/2))} \leq \Psi(k^{-1})$$

<u>Pf of claim</u> Let $\varphi = \sum |z_i|^2$ on $B(0,R)$

since $J_k \to J_{Euc}$ in $C^{1,\alpha}$

$$\sqrt{-1}\, \partial_{J_k} \bar\partial_{J_k} \varphi > \frac{1}{2} \omega_k \quad \text{for } k \gg 1$$

$\Rightarrow$ can apply Hörmander estimate to L = trivial bundle over $(B(0,R), J_k, \omega_k)$ with metric $e^{-\varphi}$

$$\cancel{\mathrm{Ric}(\omega_k)} + \sqrt{-1}\Omega > \frac{1}{4}\omega_k$$

~~we have trivia~~

Let $\sigma_0 = \partial_{J_k} z_1 \wedge \dots \wedge \partial_{J_k} z_n \in \Omega^{n,0}(L)$

$\sigma_i = z_i \sigma_0$ for $i = 1, \dots, n$

$$\|\bar\partial_{J_k} \sigma_i\|_{C^\alpha} < \Psi(k^{-1})$$

$\Rightarrow$ can find $u_i \in \Omega^{n,0}(L)$ s.t. $\bar\partial_{J_k} u_i = \bar\partial_{J_k} \sigma_i$

and $\|u_i\|_{L^2} \leq 2\|\bar\partial_{J_k}\sigma_i\|_{L^2} < \Psi(k^{-1})$

So $s_i = \sigma_i - u_i$ are holo.

$\Rightarrow \|\bar\partial_{J_k} u_i\|_{C^\alpha} < \Psi(k^{-1})$

Elliptic estimates $\Rightarrow \|u_i\|_{C^{2,\alpha}(B_{R/2})} < \Psi(k^{-1})$

so $\|\sigma_i - s_i\|_{C^{2,\alpha}(B_{R/2})} < \Psi(k^{-1})$

$\Rightarrow$ for large k, on $B_{R/2}$

$z_i^{(k)} = \frac{s_i}{s_0}$ are holo. coords.

Since L_k have curvature $-\sqrt{-1}\,\omega_k$

and $\|\sqrt{-1}\,\partial_{J_k}\bar\partial_{J_k}|z^{(k)}|^2 - \omega_k\|_{C^\alpha} < \Psi(k^{-1})$

$\Rightarrow$ can find holo. section s_k of L_k over $B_{R/2}$

s.t. $\left||s_k|^2 - e^{-|z^{(k)}|^2}\right| < \Psi(k^{-1})$

<u>Case of general $C(X)$</u>

$(M_k, p_k, g_k) \xrightarrow[\text{pointed GH.}]{} (C(X), x^*)$

Fix $R>0$, $\rho>0$. $\mathcal{R}, S \subset C(X)$ regular & singular sets.

$\mathcal{R}$ admit Ricci flat cone metric

<u>Claim</u> $\exists$ cutoff fn. ψ_ρ s.t.

$$\psi_\rho = \begin{cases} 0 & \text{in } B_{\rho^2}(S) \quad (\rho^2\text{-nhood of } S) \\ 1 & \text{outside of } B_\rho(S) \end{cases}$$

and $\int_{C(X)} |d\psi_\rho|^2 e^{-d(x^*,\cdot)^2} < \Psi(\rho|n,v)$

<u>Pf. of claim</u> uses result of Cheeger-Jiang-Naber:

$$\mathrm{Vol}\,(B_t(S) \cap B_1(q)) \le C(n,v)\, t^2 \quad \forall q \in C(X)$$

Let $\chi(s) = \begin{cases} 1 & \text{if } s<1 \\ 0 & \text{if } s>2 \end{cases}$

and set $\psi_\rho = \chi\left(\dfrac{\ln d(S,\cdot)}{\ln \rho}\right)$

<u>Calculate</u>: $\int_{C(X)} |d\psi_\rho|^2 e^{-d(x^*,\cdot)^2} < \dfrac{C(n,v)}{|\ln \rho|}$

$\mathrm{Supp}(\psi_\rho) \subset \mathcal{R}$, i.e. given $\varepsilon_0 > 0$, $\forall q \in \mathrm{Supp}(\psi_\rho)$

we have $d_{GH}(B_r(q), B_r(0)) < \varepsilon_0 r$

if r is suff small $(0 \in \mathbb{R}^{2n})$

$\Rightarrow$ we can cover $\mathrm{Supp}(\psi_\rho) \cap B_R(x^*)$

with finite collection $B(q_i, r_i)$ s.t.

$$d_{GH}\left(B(q_i, \varepsilon_0^{-1} r_i), B(0, \varepsilon_0^{-1} r_i)\right) < \varepsilon_0 r_i$$

Let $q_i^{(k)} \in M_k$ s.t. $q_i^{(k)} \to q_i$

Previous special case $\Rightarrow$ we can find

holo. charts $\{z_{i1}^{(k)}, \dots, z_{in}^{(k)}\}$ on $B(q_i^{(k)}, r_i)$

and holo. sections $s_i^{(k)}$ of L_k

s.t. $\frac{1}{10} < |s_i^{(k)}| < 10$, for $k \gg 1$

transition functions are uniformly bounded

$\Rightarrow$ we can assume that they converge

so the charts $\{z_{i1}^{(k)}, \dots, z_{in}^{(k)}\}$ converge

to charts $\{z_{i1}, \dots, z_{in}\}$ on $B(q_i, r_i)$

& $s_i^{(k)}$ converge to sections of limiting

Hermitian line bundle (L, h) on $\mathrm{supp}(\psi_f) \cap B_R(x^*)$

On $B(q_i, r_i)$ we have holo. section

s_i of L. Curvature $\sqrt{-1}\,\Omega_L = \omega_{\mathcal{R}}$

Ricci flat Kähler cone metric on $\mathcal{R}$

$C(X)$ is a cone, so $\omega_{\mathcal{R}} = \sqrt{-1}\,\partial\bar\partial\, d(x^*, \cdot)^2$

Let $L' = L$ with metric $h' = h e^{d(x^*, \cdot)^2}$

so $\sqrt{-1}\,\Omega_{L'} = \omega_{\mathcal{R}} - \sqrt{-1}\,\partial\bar\partial\, d(x^*, \cdot)^2 = 0$

i.e. L' is a flat line bundle on $\mathrm{supp}(\psi_f) \cap B_R(x^*)$

If L' were trivial, we could define constant section σ_0' of L'

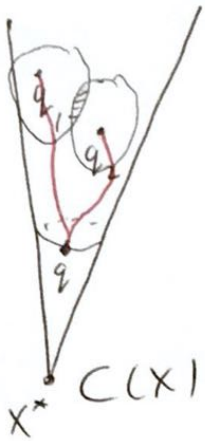
$\rightsquigarrow$ holo. section σ_0 of L with

$$|\sigma_0|_h^2 = e^{-d(x^*, \cdot)^2}$$

<u>Problem:</u> L' may have holonomy

Fix basepoint $q \in \partial B_1(x^*) \cap \mathrm{supp}(\psi_\rho)$

and paths joining q to q_i $\left(\begin{array}{c}\text{can assume} \\ \mathrm{supp}(\psi_\rho) \cap B_R(x^*) \\ \text{is connected}\end{array}\right)$



use parallel transport to define constant sections σ_i' of L' over each $B(q_i, r_i)$ s.t. $|\sigma_i'| = 1$

We have finite collection of overlaps where $\dfrac{\sigma_i'}{\sigma_j'} \in S^1 \subset \mathbb{C}$

<u>Lemma</u> Given $\delta_1 > 0$ and $\{\lambda_1, \dots, \lambda_N\} \subset S^1$, $\exists K = K(\delta_1, N)$ & $m < K$ s.t. $|\lambda_i^m - 1| < \delta_1 \quad \forall i$